

Semantic Communication for Edge AI

Can a machine communicate what matters for a task—without sending every pixel perfectly?



THESIS

Train the sender and receiver together so the information needed for classification survives the link—then compare it fairly with a strong conventional system.

A simulation-first capstone with an optional real-radio demonstration later.

The AI is only as useful as the link feeding it

Imagine a remote camera sending images to a server that decides what it sees.

REAL-WORLD TENSION

- 1** The sensor is constrained Bandwidth, latency and energy are limited at the edge.
- 2** The wireless link is imperfect Noise can corrupt a short message or make decoding fail.
- 3** The receiver has a narrow goal For classification, the final question is often simply: what is in the image?

DESIGN QUESTION

Why spend the entire link budget reconstructing every pixel if the receiver ultimately needs a task decision?

The project tests this question; it does not assume the learned method must win.

Goal: preserve classification performance when communication resources are scarce and the link becomes noisy.

From bit-perfect delivery to task success

CONVENTIONAL COMMUNICATION Compress the image → protect the bits → reconstruct the file → classify it

Primary objective: recover source bits and pixels

TASK-ORIENTED COMMUNICATION Learn a representation → send it through noise → recover what the task needs

Primary objective: retain information useful for classification

EXPECTED SIGNATURE

As the link worsens, a separated chain may hit a decoding cliff; a learned joint system may degrade more gradually.

IMPORTANT CONTROL

A digital system can also send learned features—so the experiment includes that third arm.

Research question: does the task-oriented joint design help after bandwidth, noise and comparison strength are matched fairly?

How prior research leads to this project

Finite blocklength

[1–4]

Rate backs off from capacity; source and channel dispersions interact.

OPEN ISSUE

Need measured practical short-packet loss.

Learned compression

[11–17]

End-to-end transforms and entropy priors improve rate–distortion.

OPEN ISSUE

Reconstruction optimization is not task optimization.

DeepJSCC

[5–10, 24]

Continuous learned mappings exhibit graceful degradation under mismatch.

OPEN ISSUE

Most image studies foreground reconstruction quality.

Task-oriented communication

[18–23]

Learned representations can preserve remote inference under link constraints.

OPEN ISSUE

Task awareness must be separated from joint coding.

Gap carried forward: three-way attribution at equal channel uses, with a strong adaptive classical comparator and paired failures retained.

What we will build—and what counts as success

OBJECTIVES

1. Build a reproducible normalized-AWGN DJSCC pipeline.
2. Implement a standards-derived, non-strawman JPEG 2000 + NR LDPC baseline.
3. Tune classical quality, code rate and modulation on validation only.
4. Add the task-aware digital feature control for attribution.
5. Evaluate paired image outcomes once on the sealed test split.
6. Report positive, null or negative outcomes under one frozen protocol.

H1 · PRIMARY

Low-SNR separation: learned – classical_adaptive

A point qualifies when $\sqrt{N} \cdot \Delta(s) / \sigma(s) > 1.96$. H1 is supported only if the longest run R_{obs} contains at least three consecutive qualifying points at or below the training SNR and the calibrated run p-value ≤ 0.05 .

Effect size of record: mean paired accuracy difference over the full low-SNR region.

H2 Graceful vs cliff

Fixed classical curve drops ≥ 30 pp while learned drops ≤ 15 pp over the frozen window.

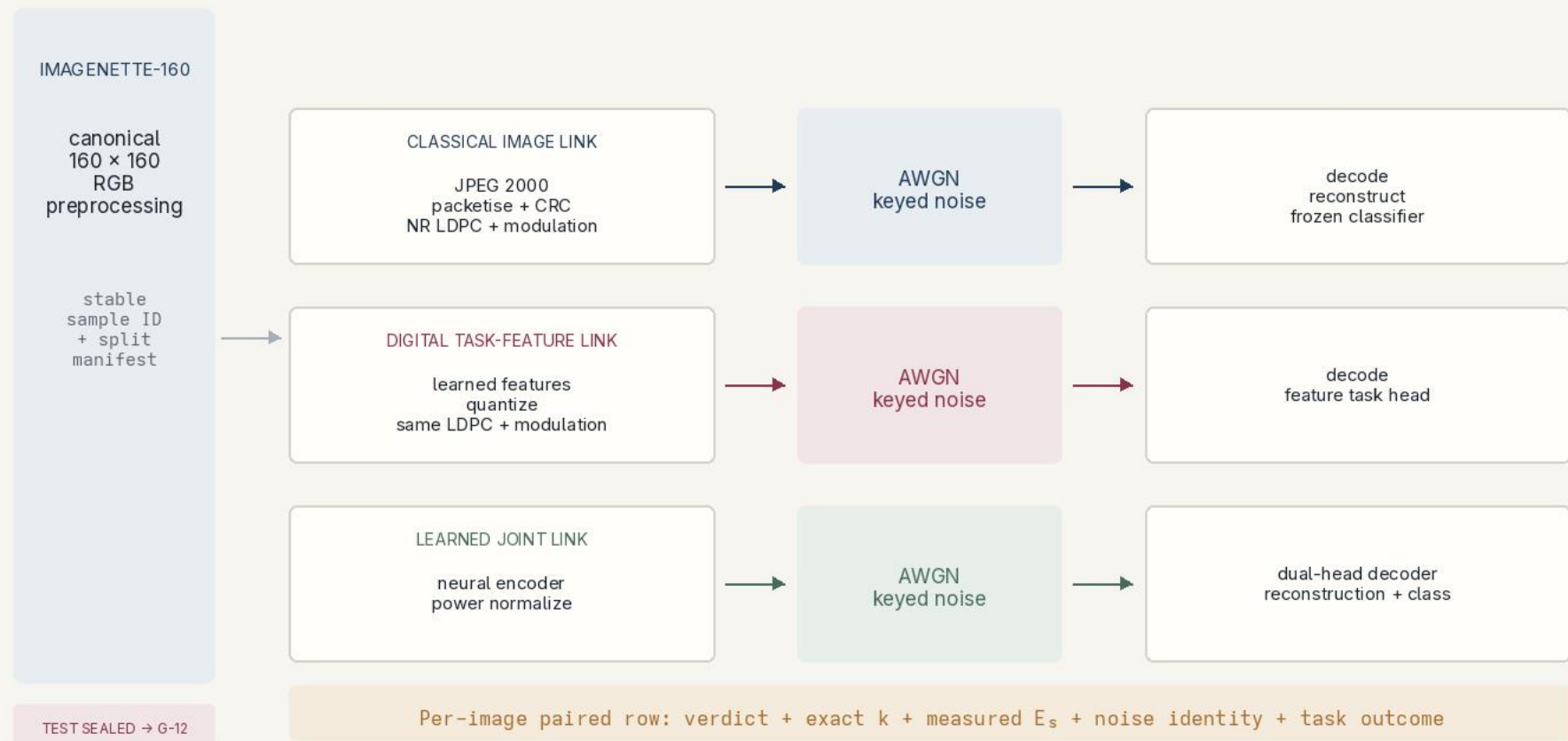
H3 Convergence

Paired gap trends toward zero across SNR; crossover is reported, not required.

H4 Attribution

Credit joint coding only if DJSCC also exceeds the task-aware digital control under the H1 rule.

The experiment in one picture



How we keep the comparison fair

Equal resource

Same complex-symbol budget k and E_s/N_0 definition; realized symbol energy is logged.

Strong comparator

JPEG 2000 quality, LDPC rate and BPSK/QPSK/16-QAM tune per SNR on validation.

Exact accounting

Container, CRC, code-block and filler bytes count; decode failures remain in the denominator.

Measured BLER

Every required physical identity is characterized; missing evidence is never treated as zero BLER.

Outage policy

Constant-class fallback is measured from validation counts—100/1,000—not assumed as 1/10.

Classifier control

Clean and artifact-finetuned classifier passes separate codec shift from link failure.

Attribution arm

Quantized learned features traverse the same digital chain, isolating task awareness from joint coding.

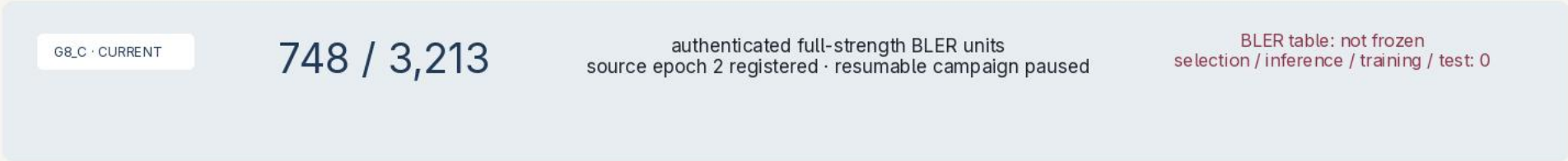
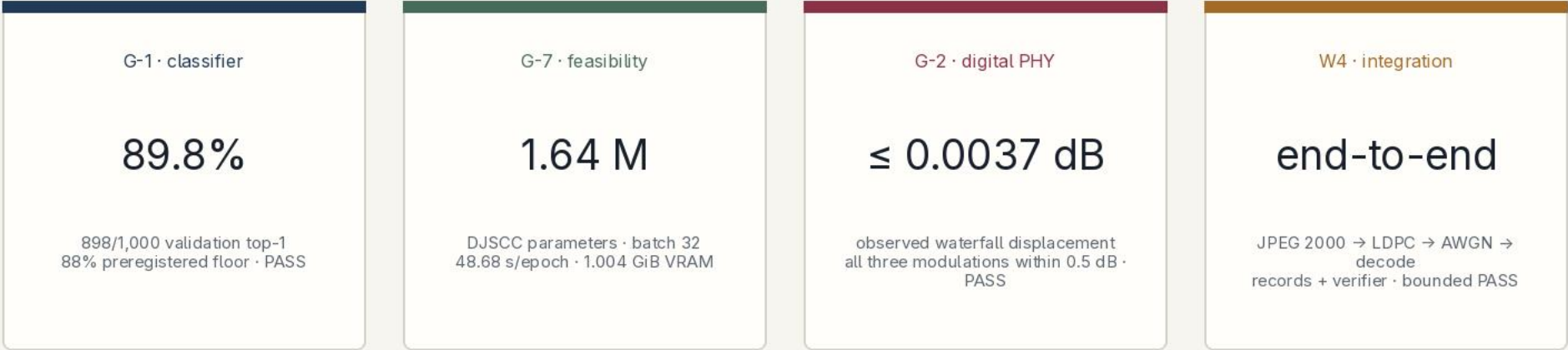
One test campaign

All choices freeze before G-12; paired inference retains complete image × system trajectories.

Fail-closed rule: uncharacterized, infeasible and failed cases are explicit outcomes—not silently dropped rows.

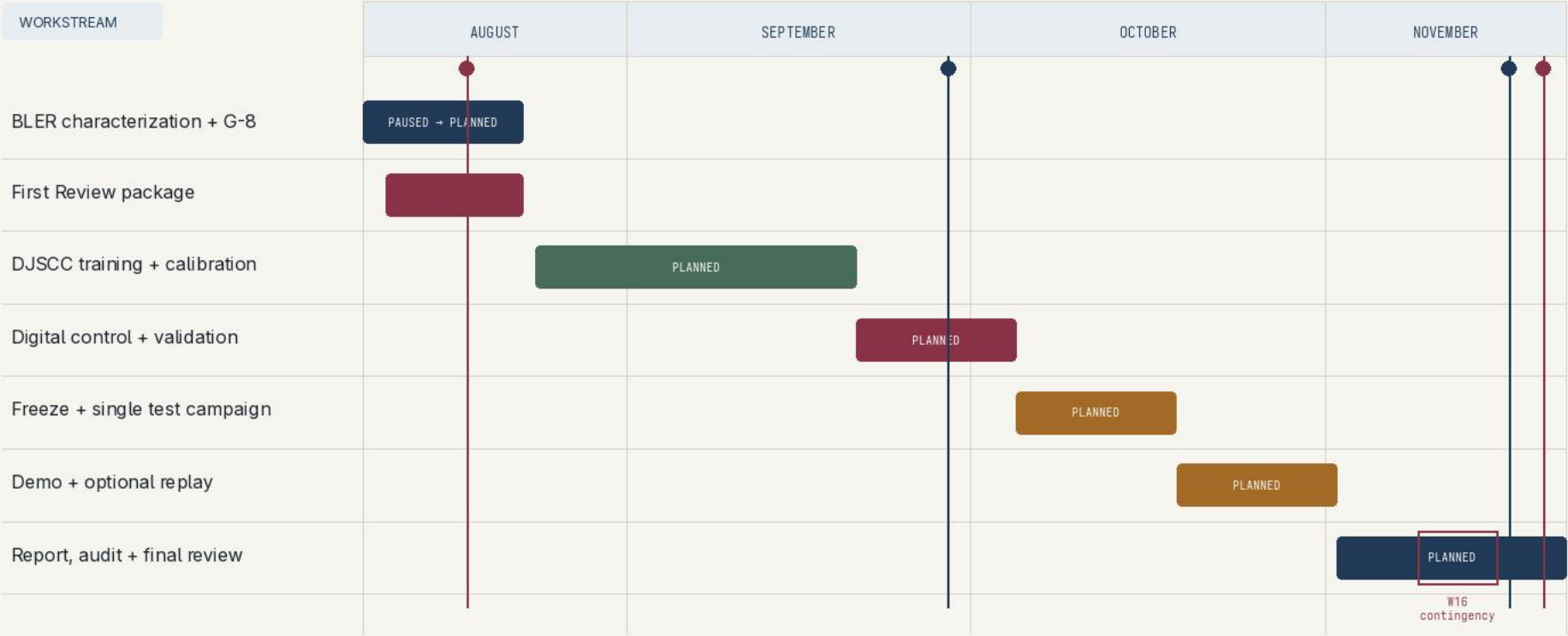
Engineering and gate evidence—not learned-vs-classical results

NO HEADLINE COMPARISON EXISTS YET · bounded smoke is integration evidence, not scientific outcome evidence



Interpretation: the reference task, computational budget and digital chain are verified; the comparison itself remains ahead.

Project Gantt chart



Review dates: 18–22 Aug · 29 Sep–3 Oct · 17–21 Nov · report due 20 Nov

W16 is report contingency—not room for new experiment scope.

What this First Review should confirm

RESEARCH SCOPE

Project boundary

One narrow task: classify an image after communication over a bandwidth-limited noisy link.

The sender runs an encoder; the receiver owns the task head. That deployment split is fixed.

Tier 1 uses a normalized simulated AWGN channel so the scientific comparison is controlled and reproducible.

The result may be positive, null or negative.

IMPLEMENTATION

Deployment ladder

TIER 1 · required
Simulation-first, offline and reproducible.

TIER 2 · stretch after G-5
Conducted SDR I/Q replay; prerecorded outcome expected.

TIER 3 · stretch
Edge placement only if latency and procurement gates pass.

Candidate: HackRF One + attenuation chain + RTL-SDR; no purchase yet.

PANEL ASK

Decision requested

Confirm that the project asks the right question and compares the systems fairly:

- image-classification is the task
- communication resources are matched
- the classical baseline is strong
- the digital feature arm supports attribution
- simulation-first Tier 1 is sufficient

Guide acknowledgement: PENDING

TECHNICAL BOUNDARY

OpenJPEG 2.5.4 + TS 38.212-derived LDPC/rate matching over abstract AWGN—not a full 5G NR link claim

No guaranteed win, measured energy saving or real-radio generalization; completion means executing the preregistered protocol.

Explicit First Review rubric-to-slide map

CRITERION	SLIDES	EVIDENCE EXPOSED IN THE DECK
Motivation	2–3	Short packets, fixed k , task accuracy, finite-blocklength costs
Objectives	3, 5–7, 10	Three-arm build, fair protocol, attribution control, decision ask
Hypothesis	5	Exact H1 run rule; H2–H4 preregistered; crossing not required
Problem Survey	2, 4, 12	30-source synthesis across four literature families
Subject Knowledge	3–8, 10	AWGN, LDPC, JPEG 2000, fairness, evidence and scope boundary
Time Plan	9	Gate-ordered critical path and fixed review/report dates

Final PASS still requires the real PPT review, four-member rehearsal, guide acknowledgement and annotated review-1-basis tag.

Selected references and evidence provenance

SELECTED LITERATURE

[1] Shannon, A Mathematical Theory of Communication, 1948.

[2] Gastpar, Rimoldi & Vetterli, To Code, or Not to Code, 2003.

[3] Polyanskiy, Poor & Verdú, Finite Blocklength, 2010.

[4] Kostina & Verdú, Lossy JSCC at Finite Blocklength, 2013.

[5] Boursoulatz, Kurka & Gündüz, DeepJSCC, 2019.

[6] Kurka & Gündüz, DeepJSCC-f, 2020.

[7] Kurka & Gündüz, Bandwidth-Agile DeepJSCC, 2021.

[11] Ballé, Laparra & Simoncelli, Learned Compression, 2017.

[12] Ballé et al., Scale Hyperprior, 2018.

[17] Blau & Michaeli, Perception-Distortion, 2018.

[18] Jankowski, Gündüz & Mikołajczyk, Wireless Retrieval, 2021.

[19] Shao, Mao & Zhang, Task-Oriented Edge Inference, 2021.

[20] Xie et al., Deep Learning Enabled Semantic Communication, 2021.

REPOSITORY EVIDENCE

G-1	results/reference_classifier/g1_adjudication.json 89.8% validation; test sealed
G-7	results/profiling/g7_djscc_profile.json 1.64M params; measured epoch/VRAM
G-2	results/baseline/g2/g2_adjudication.json golden vectors + BLER conformance
W4	results/baseline/w4/integration_adjudication.json bounded end-to-end integration
G8_C	results/baseline/g8/campaign_state.json authenticated characterization cursor
Spec	spec/SPEC.md normative hypotheses and requirements
Plan	docs/gantt-plan.md dates, gates and current status
Deploy	docs/deployment-dossier.md simulation-first + SDR stretch

Refresh G8_C status from instructions/RESUME.md on submission day.